

Einstein-QED: variables, cutoff matching and two verified reduced closures

Research snapshot: 9 September 2026. Round 7 of the Physics Observatory.

What changed and what this establishes

This study makes a proposed additional variable concrete: a canonical scalar field with a specified gauge-kinetic interaction, its own equation and its own energy. Two distinct reduced models were derived and simulated. In the finite quantum-mode model, scalar exchange closes the exact energy balance. In the classical Einstein-Maxwell-scalar model, action-derived directional stress preserves the gravitational constraint. These are useful mathematical and computational results under declared assumptions. Their novelty in the literature has not been established.

The full Einstein-QED theory has not been completed. The two models do not yet share a quantum state, current, directional pressures or covariant subtraction. Adding a variable that keeps a separated coefficient positive does not by itself establish those missing quantities. The project therefore records a successful finite closure and a successful classical-gravity closure alongside an explicit unfinished interface.

The advisor and skeptic supplied independently derived equations. Both implementation drafts were then audited and corrected by the root after the delegated agents reached a usage limit. The source files, original drafts, raw histories and validation records are retained. This report provides derivations, assumptions, evidence and decisions rather than a claim to reproduce private internal reasoning.

1. Resolve the ambiguity in the Landau cutoff

N labels retained Landau levels. K bounds canonical longitudinal momentum. nK resolves an integral at fixed N,K. None is automatically a physical renormalization scale. A continuous interpolation of an analytic digamma expression does not create fractional Landau states or a new degree of freedom. The scalar mass ratio ν in the finite solver is also distinct from the interpolation symbol used in the variable inventory.

For one Dirac species, the high-energy one-loop QED equation is $\mu_{RG} \, d\epsilon/d\mu_{RG} = e^3/(12 \pi^2) + O(e^5)$. Integrating the displayed term yields $1/e^2(\mu_{RG}) = 1/e^2(\mu_0) - \log(\mu_{RG}/\mu_0)/(6 \pi^2)$. Its formal pole is an extrapolation beyond perturbative control. It is not identified here with loss of positivity of the finite-mode divided coefficient. See the exact-RG and charge-renormalization sources in the reference ledger.

The finite coefficient is $Z = 1 + \chi - e^2 C$. The previously proved continuous-integral maximum at $a=0$ gives a constructive possibility: set $\chi_{NK} = z_{ref} - 1 + e^2 C_{NK}(0)$, constant in time for each run. Then $Z_{NK}(a) = z_{ref} + e^2 [C_{NK}(0) - C_{NK}(a)] \geq z_{ref} > 0$. This is valid algebra and changes the fixed- χ premise. It does not prove that this is an allowed universal physical counterterm or that the combined current and stress converge.

Conversely, for bounded potential and a uniformly bounded replacement coefficient h , the shifted window contains the centered window $K-A$. Its positive coefficient diverges along cofinal N, K . Therefore $Z = h - e^2 C$ cannot retain a uniform positive gap. A scalar confined to a common compact interval with a continuous f is bounded and cannot evade this result. A regulator-dependent

divergent scalar, singular f or running charge changes the assumptions and creates new physical matching obligations.

The exact integral and arbitrary quadrature must remain distinct. An independent two-node Gauss counterexample produces a larger coefficient near a retained node than at the origin. The reference-at-zero inequality does not transfer to that discretization without a separate bound.

2. Introduce a variable with a specified action

The organizing covariant action is Einstein gravity plus a canonical scalar, potential and gauge kinetic term: $S = \int \sqrt{-g} [M_{\text{Pl}}^2 (R - 2\Lambda)/2 - (\text{grad } \Phi)^2/2 - V(\Phi) - f(\Phi) F^2/4]$. A genuine quantum extension also needs one causal state-dependent in-in functional and consistent effective-action terms. Simply writing that functional as a symbol is not computing it.

Scalar-dependent electromagnetic couplings are an established theory class, with precedents in Bekenstein, dilaton gravity and anisotropic inflation. We use them as explicit model hypotheses, not evidence that an undiscovered scalar is required by QED. The inflation reference writes the entire kinetic multiplier as a squared function; our f is already the entire multiplier. This convention matters for factors of two.

Prescribing $\chi(t)$ is a different operation. Inserting it into the old Maxwell equation gives a positive half χ -prime work defect; varying an explicitly time-dependent kinetic action instead adds a Maxwell term and gives the opposite sector-work sign. Neither gives a closed unsupported energy budget. A dynamical scalar supplies an equation and the compensating sector energy.

3. Finite quantum-mode scalar model: exact formulation

Use $s=mt$, $a=e A_z/m$, $x=e E_z/m^2$, $b=|eB|/m^2$ and $\phi=e \Phi/m$. At fixed positive masses M_i , weights w_i and canonical nodes k_i , define $p_i=k_i-a$, $h_i=(M_i, 0, p_i)$, $\omega_i=\sqrt{M_i^2+p_i^2}$. Bloch vectors retain their complete coherent evolution. S , C , D and U are the finite sums defined in Appendix A; they must not be replaced by an instantaneous particle-number ansatz.

Choose $f=1+g^2 \phi^2$, $V=nu^2 \phi^2/2$, $Z=f+\chi_b-e^2 C$. The equations are $a'=-x$, $r_i'=2 h_i \text{ cross } r_i$, $\phi'=y$, $y'=-nu^2 \phi+f_{\phi}(x^2-b^2)/2$, and $Z'x'=F-e^2(S+D x^2)-f_{\phi} y x$. The magnetic field is fixed in this reduction, but its scalar-dependent energy cannot be discarded.

The finite action includes the normalized spinor Berry term, subtracted mode energy, $Z a'^2/2$, scalar kinetic and potential terms, magnetic energy $f b^2/2$ and source term $-a F$. Varying it fixes all exchange signs. The complete first-order action and normalizations are given in Appendix A.

The shifted energy is $\tilde{W}=Z x^2/2+e^2 U+y^2/2+(nu^2+g^2 b^2)\phi^2/2$. Precession gives $U'=xS$; coefficient differentiation gives $Z'=f_{\phi} y+2e^2 Dx$. Field and mode work equals $x F-f_{\phi} y x^2/2$; scalar plus shifted magnetic work equals the omitted half. Thus $\tilde{W}'=x F$ exactly.

At $b=10, K=20$, the old exact positive-quadrature certificate gives a global gap at least $3/4$ for every finite inclusive N . Since $f \geq 1$, it remains a gap for the new model. Physical Bloch norms make U nonnegative. An epsilon-regularized square-root energy estimate bounds all state variables on every finite interval. If $\Omega^2=nu^2+g^2 b^2$ vanishes, $g=nu=0$ and the scalar is affine; it is handled separately. Smoothness and the finite-dimensional continuation criterion yield a unique global forward solution for each finite model. No regulator-removal or current-limit theorem follows.

4. Finite simulations and falsification

The recorded main preparation is $b=10$, $N=1$ inclusive, $K=20$, $nK=128$, $g=0.1$, $\nu=0.5$, $\phi_0=0.4$ and $y_0=0.1$. A normalized smooth compact drive has amplitude 0.15 and support $0 \leq s \leq 4$; the run ends at $s=6$. A delayed probe is a separate recorded source history. DOP853 uses $\text{rtol}=2\text{e-}10$ and $\text{atol}=2\text{e-}12$, with maximum step 0.05. There are 121 output samples; all quoted maxima are sampled diagnostics, not interval enclosures.

Test	Recorded result	Interpretation
Correct main energy residual	1.528e-15	Sampled work identity in this finite run
Minimum sampled kinetic coefficient	0.997329398	Numerical domain check; exact lower bound is separate
Wrong-model unaccounted energy	1.907e-05	Removing only the Maxwell exchange is distinguishable
Wrong-model predicted-defect residual	8.781e-16	Independently integrated defect explains the failure
128-to-256-node field/potential discrepancy	4.135e-07	Quadrature refinement at fixed N,K
Largest independent spinor/Bloch macro discrepancy	3.216e-11	Distinct state representation and RK45 integrator
Historical $g=0$ field discrepancy	3.719e-15	Actual archived solver was imported and compared

The zero-drive oscillator, zero-frequency affine scalar, zero-scalar invariant subspace and canonical gauge translation also pass. The delayed-source response is consistent across three centered perturbation sizes and has a negligible pre-source derivative. Its differences are already very small; this run does not demonstrate a truncation-dominated second-order convergence interval. A nonzero response alone was not accepted as an accuracy check.

5. Classical gravity model: complete directional stress

Choose the axial metric $ds^2 = -dt^2 + A^2(dx^2 + dy^2) + C \text{metric}^2 dz^2$ with aligned orthonormal E and B . Set $\tau = \mu t$, $\varphi = \Phi/M\mu$, $h = H/\mu$ and electromagnetic fields $E_{\text{cal}} = E/(\mu M\mu)$, B_{cal} likewise. Here μ is a fixed frequency, not a renormalization scale. Use $f = \exp(2c \varphi)$, potential $U = \hat{m} \tau^2 \varphi^2/2$ and $\theta = 2h_{\text{perp}} + h_{\text{parallel}}$.

The energy and directional pressures are $\rho = v^2/2 + U + \rho_{\text{EM}}$, $p_{\text{perp}} = v^2/2 - U + \rho_{\text{EM}}$, $p_{\text{parallel}} = v^2/2 - U - \rho_{\text{EM}}$, with $\rho_{\text{EM}} = f(E_{\text{cal}}^2 + B_{\text{cal}}^2)/2$. Evolving E_{cal} , B_{cal} , φ , v and both spatial Einstein equations gives the complete eight-variable system in Appendix A. Initial isotropic expansion rates satisfy the Hamiltonian constraint; anisotropic stress can subsequently make the rates different.

Define $C_g = h_{\text{perp}}^2 + 2h_{\text{perp}} h_{\text{parallel}} - \rho - \lambda$. Independent differentiation gives $C_g' = -\theta C_g - Q_{\text{total}}$, where Q_{total} is the anisotropic matter-conservation residual. The scalar and electromagnetic exchange terms cancel exactly, so a satisfied constraint propagates on the regular interval. The code does not enforce this by resetting a state variable.

The source-free flux laws eliminate E_{cal} and B_{cal} in a second integration. Its gravitational RHS is shared with the production implementation, so this is a useful consistency comparison rather than wholly independent code. The skeptic separately derived the equations and implemented another

system; the root independently checked the finite model with complex spinors.

6. Gravity experiments and negative controls

Eight recorded cases cover mixed, pure electric, pure magnetic, constant-coupling, Minkowski, de Sitter, scalar-only and short contracting data. DOP853 and Radau integrate full, flux-reduced and deliberately incomplete equations. Two analytic fixtures are actually executed: de Sitter and a massless isotropic scalar. Additional root holdouts cover negative coupling and equal initial E,B. Equal initial fields do not imply the electromagnetic invariant remains zero when f evolves.

For the mixed case the maximum correct constraint residual is $5.551e-16$, while deleting the scalar electromagnetic force produces $3.022e-05$. The independently integrated weighted Ward defect explains that drift with residual $7.304e-16$. Across all eight cases the largest full-method discrepancy is $5.074e-12$. Analytic de Sitter and massless-scalar discrepancies are $5.551e-17$ and $3.600e-12$.

These finite-time results do not prove absence of later collapse or singularity formation. They do not contain Dirac vacuum polarization, Schwinger current or quantum pressure. The elementary scalar-gravity benchmark must not be relabeled Einstein-QED because its constraints pass.

7. Bidirectional linking and the exact remaining gap

The executable finite Horn graph contains a 17-rule route from the finite action/domain to global finite-model continuation and an 11-rule route to classical-gravity constraint propagation. Two four-rule routes certify the reference-subtraction lemma and bounded-addition obstruction. A hash guard binds the proof notes and code. Independent forward uniform-cost search certifies route cost; the bidirectional heuristic is $h=0$.

Withdrawing the Maxwell-equation certificate prevents the finite continuation route. The full semiclassical target is also underivable in the reviewed finite library. This is a useful failure: the forward results do not meet the backward obligations for common quantum stress, state, matching and evolution. It is not a proof that no mathematical solution exists elsewhere. The search checks inference replay, not the mathematical truth of its prose certificates.

The next concrete calculation is a common causal quantum current and stress in a prescribed smooth axial geometry. Derive state evolution, current, energy and both pressures together; fix allowed counterterms and one physical normalization. Independently evaluate their force Ward residual. Only then attempt self-consistent geometry and regulator limits. Reconstructing a pressure from the Ward identity being tested would make that test circular.

8. Relationship to the wider frontier

The variable extension supplies a tested exchange pattern for later strong-field backreaction. It does not settle the Festina Lente/WGC inequalities, near-extremal decay, Schwinger creation at electron-scale curvature, Cauchy-horizon mass inflation or quantum-corrected photon-graviton conversion. Those previously mapped problems retain distinct state, geometry and approximation obligations.

A low-energy Euler-Heisenberg benchmark is planned in its controlled overlap regime. Its leading polynomial does not apply at B much greater than the QED critical field. The full constant-field one-loop action has a broader amplitude range, but that fact does not supply the needed nonlocal in-in stress in a rapidly changing curved background. The 2026 de Sitter study and the July 2026 Einstein-Euler-Heisenberg preprint are relevant recent context in the source ledger, not empirical or mathematical verification of this project's scalar hypothesis.

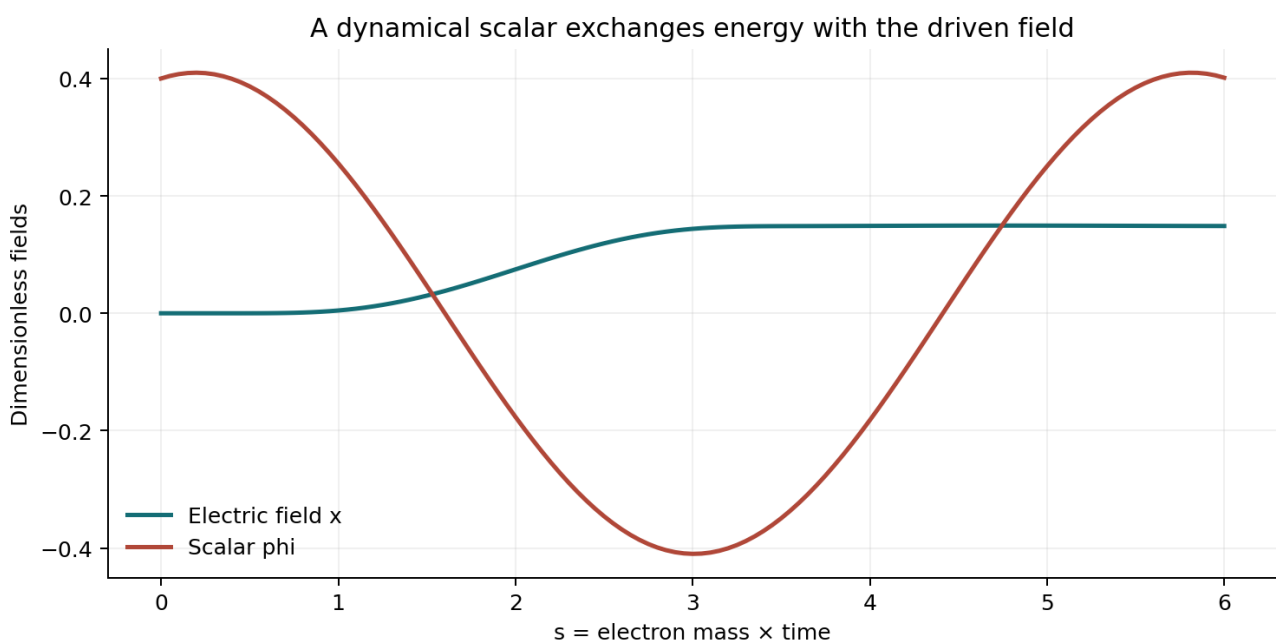
9. Validation, use and reproducibility

The independent skeptical script passes 42 checks; the root validation passes 47. Both also pass with optimized Python, where removable assertions cannot serve as acceptance gates. These counts combine different checks and are not confidence percentages. The audit preserves thirteen classes of implementation or interpretation issues, including a skipped comparator, unexecuted fixtures and clipped diagnostics.

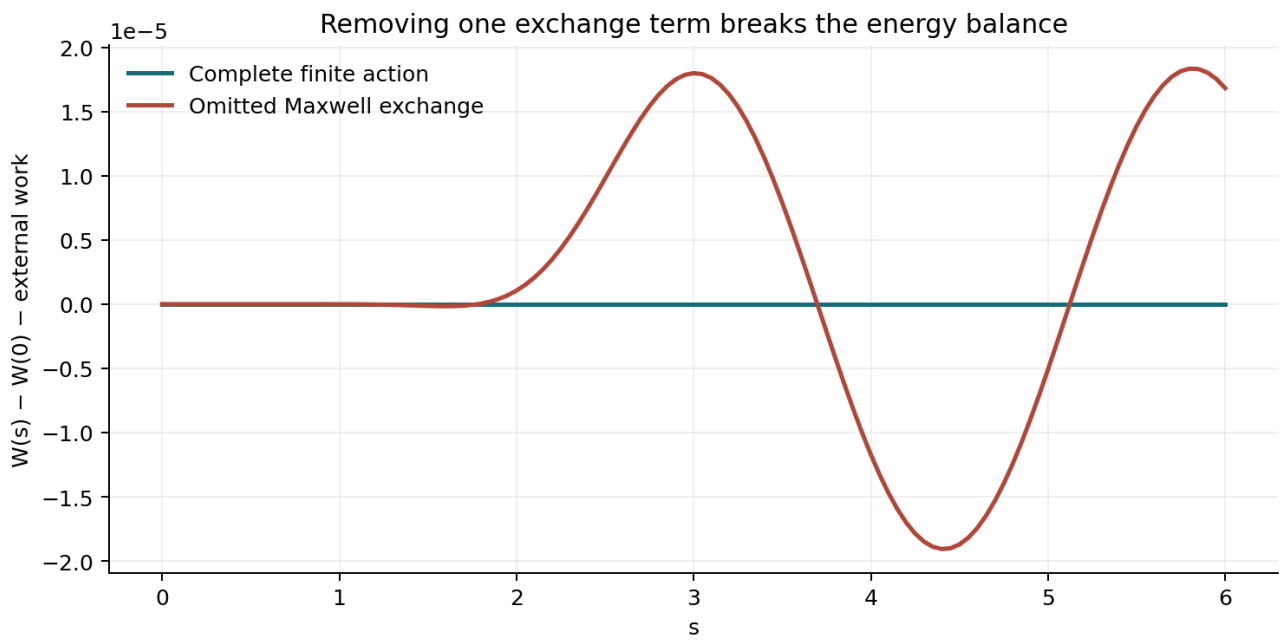
Use the research home for the current and historical summaries. The variables page explains each kind of freedom. The finite and gravity pages expose equations, recorded plots and wrong-model controls. The proof page shows all reviewed substeps and the underivable branches. Documents open on request. Downloadable CSVs are exact plotted histories; the archive contains source, primary-reference metadata, complete equation notes and this report.

To reproduce, follow README.md and requirements.txt. Do not edit the historical rounds to make a comparison pass. Intentionally changed physical parameters require fresh results, and intentionally changed proof premises require a newly reviewed manifest. The sources below support their stated framework or result; none is presented as a blanket certificate of this project.

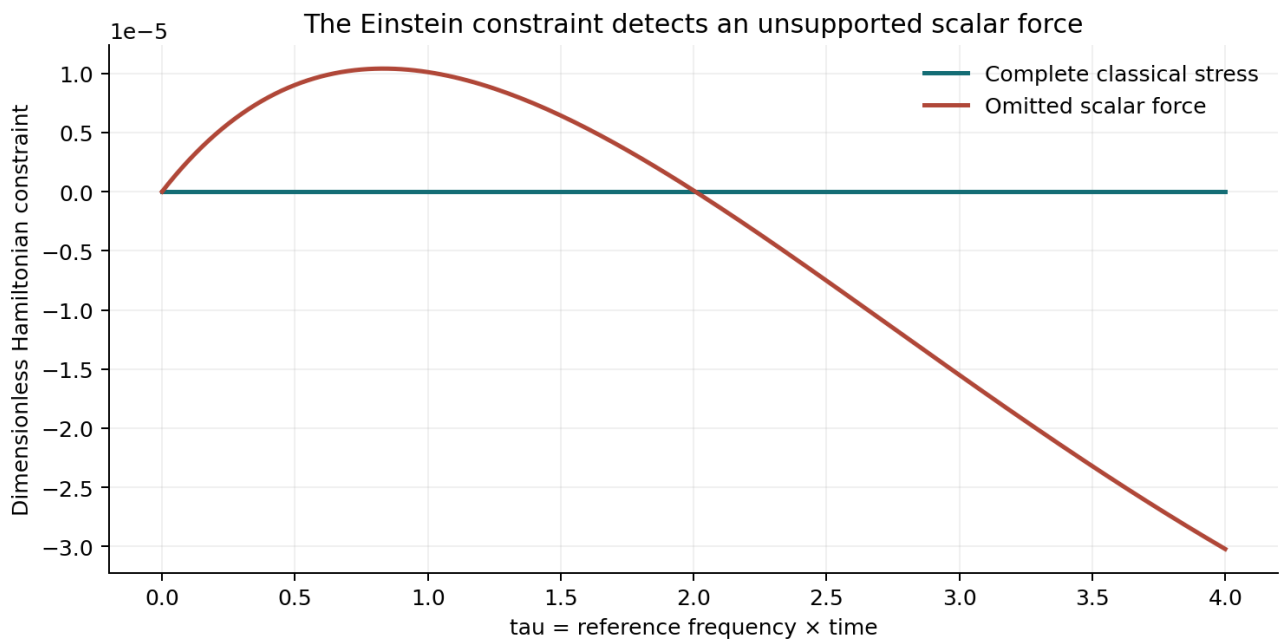
Recorded simulations



Recorded finite model: $b=10$, $N=1$ inclusive, $K=20$, $nK=128$; $g=0.1$, $\nu=0.5$, $\phi(0)=0.4$, $y(0)=0.1$. The drive has support $0 < s < 4$. These fields use electron-based units.



Correct and deliberately incomplete finite equations, with identical preparation and drive. The omitted Maxwell term produces the independently integrated defect $f_{\phi} y x^2$. The difference is physical inconsistency of the modified model, not evidence of numerical instability.



Classical Bianchi I mixed-field case, $c=0.45$, $m_{\text{hat}}=0.35$, $\lambda=0.01$. Correct evolution retains the constraint; removing only the scalar electromagnetic force creates the predicted Ward-driven drift. No QED quantum stress is present.

Appendix A - full action and equation contract

Advisor derivation, 9 September 2026; contract version 1.1. This document defines two new, distinct mathematical models. They are linked by their variational gauge-kinetic coupling and work identity; they are **not** a closed semiclassical Einstein-QED system. Numerical outcomes belong to the separately executed experiment records. The equations below are a calculation contract, not a claim that its simulations have already passed. Version 1.1 makes the finite scalar's canonical rescaling explicit; it changes none of the differential equations.

1. What is allowed to vary

Symbol	Type	What changes	What it cannot establish alone
N,K	Ultraviolet regulator labels	Which Landau modes and canonical longitudinal momenta are retained	Physical time evolution or charge running
chi_(N,K)	Regulator-dependent matching coefficient	A member of a family of finite actions	Renormalized current/stress convergence
mu_RG	Renormalization scale	The bookkeeping scale of renormalized parameters, with compensated observable dependence	A new spacetime degree of freedom
chi(t)	Prescribed time-dependent coefficient	A driven kinetic sector and an additional work exchange	Conservation without a support sector
phi(t)	Dynamical scalar	A new state variable with an equation, energy and stress	A QED ultraviolet completion or observed new particle
f(phi)	Positive gauge-kinetic function	A specified interaction in a specified action	Permission to retune the coupling after a failed run

The symbol mu in the gravity scaling below is a fixed reference frequency, not mu_RG.

1.1 Reference subtraction is valid algebra, with an unclosed physical interpretation

For the round-6 exact longitudinal integral, fix a positive target z_ref and define, at each finite regulator pair,

$$\text{chi}_{(N,K)} = z_{\text{ref}} - 1 + e^2 C_{(N,K)}(0).$$

Then

$$Z_{(N,K)}(a) = z_{\text{ref}} + e^2 [C_{(N,K)}(0) - C_{(N,K)}(a)] \geq z_{\text{ref}}.$$

The inequality follows from the proved continuous-integral maximum at a=0. It does not automatically hold for a finite arbitrary quadrature, and it is not a theorem about a common renormalized quantum current. The finite matching coefficient diverges along the joint cutoff family. A physical interpretation needs one regulated action, a physical charge normalization, the associated

common subtraction in current and stress, and estimates on the state-dependent terms. Across this family N, K are labels; no $d\chi/dt$ term arises unless a separate time-dependent prescription is introduced.

Conversely, if a proposed replacement only adds a uniformly bounded amount to the kinetic coefficient, it cannot cancel the established cofinal divergence. For $|a_-(N, K)| \leq A$, the shifted window contains $[-K+A, K-A]$ when $K > A$, so

$$C_-(N, K)(a_-(N, K)) \geq C_-(N, K-A)(0) \rightarrow \text{infinity}.$$

Thus bounded $f(\phi) + \chi$ cannot produce a uniform positive denominator. A common compact scalar range and continuous f are sufficient for that boundedness. Allowing the scalar state to diverge with the regulator changes the question and does not supply a physical matching condition.

1.2 Two different time-dependent substitutions have different work defects

Keep the original finite-mode definitions and replace only χ by $\chi(t)$. If the old divided Maxwell equation is otherwise left unchanged, the previously defined electromagnetic/mode energy obeys

$$W' = x F + (\chi'/2) x^2.$$

If instead the time-dependent coefficient is derived from a kinetic term in an action, the Maxwell numerator gains $-\chi' x$, and the same sector energy obeys

$$W' = x F - (\chi'/2) x^2.$$

These are different equations. Neither supports a claim of a conserved closed system until the driving sector and its energy exchange are specified. The sign cannot be selected after inspecting a plot. The dynamical model below supplies that missing sector explicitly.

2. Model F: a finite variational scalar-Maxwell-Bloch extension

F1. Domain and definitions

Keep the exact finite regulator, fixed $b > 0$, fixed canonical nodes k_i , positive weights w_i , positive masses M_i , fixed coupling $e^2 > 0$ and fixed matched χ_b . The round-6 certificate applies at $b=10, K=20$ to every finite inclusive Landau cutoff and exact positive per-level quadrature weights summing to $2K$. It does not certify rounded arbitrary inputs or a continuum limit.

Use the existing electron-scaled time and electromagnetic variables $s=mt$, $a=e A_z/m$, $x=e E_z/m^2$, $b=|eB|/m^2$, with $e=\sqrt{e^2} > 0$. A canonical physical scalar Φ can be assigned by $\phi=e \Phi/m$; the natural reduced energy-density unit is m^4/e^2 . Then $nu=m_\Phi/m$, and $f(\Phi)=1+(g e \Phi/m)^2$. This rescaling derives the scalar kinetic and magnetic energy prefactors below; it does not derive the entire regulated quantum action from a covariant theory. The gravity branch instead uses a Planck-scaled scalar.

For each retained mode set

$$p_i = k_i - a, h_i = (M_i, 0, p_i), \omega_i = \sqrt{M_i^2 + p_i^2}.$$

$$S = \sum_i w_i (r_{iz} + p_i / \omega_i).$$

$$C = \sum_i w_i M_i^2 / (4 \omega_i^5).$$

$$D = \sum_i w_i 5 M_i^2 p_i / (8 \omega_i^7).$$

$$U = \sum_i w_i (h_i \cdot r_i + \omega_i).$$

Choose constant parameters g and $nu \geq 0$, define

$$f(\phi)=1+g^2 \phi^2, f_\phi=2g^2 \phi, V(\phi)=\nu^2 \phi^2/2,$$

$$Z(a,\phi)=f(\phi)+\chi_b e^2 C(a).$$

The scalar gauge coupling is additional model physics. It is not the regulator-dependent matching coefficient of section 1 and is held fixed under a regulator comparison.

F2. Complete coupled system

$$a'=-x.$$

$$r_i'=2 h_i \text{ cross } r_i.$$

$$\phi'=y.$$

$$y'=-\nu^2 \phi + (f_\phi/2)(x^2-b^2).$$

$$Z x'=F(s)-e^2(S+D x^2)-f_\phi y x.$$

The background magnetic field is held fixed in this homogeneous, nonexpanding reduction. There is no evolving metric or horizon. No external electric source is hidden: $F(s)$ is the same explicitly supplied smooth drive used by the finite reference model.

F3. Variational check

For normalized two-component mode spinors z_i with $r_i=z_i^\dagger \sigma z_i$, use the finite action with Lagrangian

$$L = Z(a,\phi) a'^2/2 + \phi'^2/2 - V(\phi) - f(\phi)b^2/2 \\ + e^2 \sum_i w_i [i(z_i^\dagger z_i' - z_i'^\dagger z_i)/2 - z_i^\dagger(h_i \text{ dot } \sigma)z_i - \omega_i] - a F(s).$$

The symmetrized Berry term differs from the usual first-order expression by a boundary term. The equations for the mode spinors reproduce Bloch precession. The source sign is fixed: the term is $-a F$, because $a'=-x$.

The identities $C_a=2D$, $Z_a=-2e^2D$ and

$$\partial_a \sum_i w_i [-z_i^\dagger(h_i \text{ dot } \sigma)z_i - \omega_i] = S$$

give the Maxwell equation in F2 by the Euler-Lagrange equation for a . The scalar equation is its own Euler-Lagrange equation. This establishes internal consistency of a declared finite hybrid action. It does not derive this action as the complete truncation of covariant quantum QED in an evolving geometry.

F4. Exact energy/work identity and continuation statement

The full source-free Hamiltonian is

$$W = Z x^2/2 + e^2 U + y^2/2 + V(\phi) + f(\phi)b^2/2.$$

Subtract the constant magnetic baseline $b^2/2$ when measuring changes:

$$W_{\text{tilde}} = Z x^2/2 + e^2 U + y^2/2 + 0\omega^2 \phi^2/2,$$

$$\text{where } 0\omega^2 = \nu^2 + g^2 b^2.$$

The complete cancellation can be checked without a solver:

1. Precession gives $(|r_i|^2)'=0$ and $U'=xS$.
2. $C'=-2Dx$, so $Z'=f_\phi y+2e^2Dx$.
3. $(Zx^2/2+e^2U)'=xF - (f_\phi y/2)x^2$.

4. $(y^2/2 + \omega^2 \phi^2/2)' = (f_\phi y/2)x^2$.

5. Therefore $W_{\text{tilde}}' = W' = xF$.

For physical Bloch norms $|r_i| \leq 1$, every term $h_i \cdot r_i + \omega_i$ is nonnegative. If the original finite denominator is globally at least $\mu_{\text{gap}} > 0$, then $f \geq 1$ retains that lower bound. For the round-6 exact-coefficient family, $\mu_{\text{gap}} = 3/4$ is available.

For continuous drive, epsilon-regularizing the square root at zero energy gives

$$\sqrt{W_{\text{tilde}}(s)} \leq \sqrt{W_{\text{tilde}}(0)} + (1/\sqrt{2 \mu_{\text{gap}}}) \int_0^s |F(u)| du.$$

Consequently

$$|x(s)| \leq \sqrt{2 W_{\text{tilde}}(0)/\mu_{\text{gap}}} + (1/\mu_{\text{gap}}) \int_0^s |F(u)| du.$$

Also $|y| \leq \sqrt{2 W_{\text{tilde}}}$ and, when $\omega > 0$, $|\phi| \leq \sqrt{2 W_{\text{tilde}}}/\omega$. The potential a is bounded on any finite interval by integrating $a' = -x$. The finite mode norms remain bounded. Smoothness of the vector field on $Z \geq \mu_{\text{gap}}$ and the standard finite-dimensional ODE continuation criterion then prove a unique global forward solution for this finite model. If $\omega = 0$, necessarily $\nu = g = 0$ at fixed $b > 0$; the scalar is decoupled and affine in time, so the same finite-time continuation conclusion holds separately.

This does not bound the current uniformly over regulator removal, prove continuum convergence, or supply quantum directional stress. It is a genuinely stronger finite-model theorem under explicitly changed equations.

F5. Analytic fixtures and discriminating falsifier

- **Zero source, stationary modes:** set $x=0$, fixed a and $r_i = -h_i/\omega_i$. Then $\phi'' + \omega^2 \phi = 0$ exactly. This fixture tests changing f with a closed analytic scalar trajectory while the electromagnetic vacuum remains stationary.
- **Decoupling:** $g=0$ returns the previous driven finite Maxwell-Bloch equations plus an independent oscillator. With matching initial data and source, electromagnetic histories must agree.
- **Degenerate scalar preparation:** $\phi(0)=y(0)=0$ leaves the scalar identically zero for any g . This must not be counted as evidence that nonzero scalar exchange was tested.
- **Nontrivial exchange:** use a compact smooth drive, nonzero scalar preparation, for example $g=0.05, \nu=0.5, b=10, \phi(0)=0.2, y(0)=0.1$. These are illustrative reduced-model inputs, not measured physical constants.
- **Wrong-model control:** remove only $-f_\phi y x$ from the Maxwell numerator, while retaining the scalar equation and all energy terms. The exact defect is then $W_{\text{tilde}}' - xF = f_\phi y x^2$. Integrating this nonzero defect must explain the broken energy balance. A test which passes the wrong system with the original zero-defect criterion is invalid.

For numerical checks, compare a primary Bloch system with independent complex spinors on the same finite modes and source. Do not share a copied energy-derivative routine as the only independent evidence. Separately refine time tolerance and momentum quadrature; label any cutoff comparisons as different finite systems. Check source hash, finite diagnostics, mode norms, minimum Z , full sampled histories, and the independently integrated work. Do not clip or project energy or norms.

3. Model G: a classical scalar-Maxwell-Einstein conservation bridge

G1. Covariant action and domain

Use natural units, signature $(-, +, +, +)$, reduced Planck mass $M_{\text{Pl}}^2 = 1/(8\pi G)$, and the curvature convention yielding the Einstein equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = M_{\text{Pl}}^{-2} T_{\mu\nu}$. The action is

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2 (R - 2\Lambda)}{2} - \frac{1}{2} (\partial_\mu \phi)^2 - V(\phi) - f(\phi) F_{\mu\nu} F^{\mu\nu} \right]$$

There is no Dirac field, quantum expectation, pair-production current or quantum counterterm in this benchmark. The use of a scalar gauge-kinetic coupling in anisotropic gravity is established in the literature; our E-and-B extension and variable conventions are derived here. The cited inflation model writes its kinetic function as f_{lit}^2 ; our f denotes the complete positive multiplier, so these conventions cannot be copied without the square. [Watanabe, Kanno and Soda, equations \(1\)-\(3\)](#).

Take axial Bianchi I

$$ds^2 = -dt^2 + A(t)^2(dx^2 + dy^2) + C_{\text{metric}}(t)^2 dz^2$$

with homogeneous parallel orthonormal E and B along z. Let $H_{\text{perp}} = \dot{A}/A$, $H_{\text{parallel}} = \dot{C}_{\text{metric}}/C_{\text{metric}}$. There is no spatial boundary or horizon. Initial data specify a homogeneous Cauchy slice with positive scale factors.

The source-free covariant equations are

$$\nabla_\mu (f F^{\mu\nu}) = 0, \quad \nabla_\mu (f F_{\nu\rho}) = 0,$$

$$\Box \phi - V_\phi - f_\phi F_{\mu\nu} F^{\mu\nu} = 0.$$

Their stress is the sum of canonical scalar stress and

$$T_{\text{EM}\mu\nu} = f(F_{\mu\alpha} F_{\nu}{}^\alpha - g_{\mu\nu} F^2/4).$$

No current or stress is added after varying the action.

G2. Dimensionless variables and complete ODEs

Choose fixed frequency $\mu > 0$. Define

$$\tau = \mu t, \quad \varphi = \phi/M_{\text{Pl}}, \quad v = d\varphi/d\tau,$$

$$h_{\text{perp}} = H_{\text{perp}}/\mu, \quad h_{\text{parallel}} = H_{\text{parallel}}/\mu,$$

$$E_{\text{cal}} = E/(\mu M_{\text{Pl}}), \quad B_{\text{cal}} = B/(\mu M_{\text{Pl}}),$$

$$\lambda = \Lambda/\mu^2, \quad U = V/(\mu^2 M_{\text{Pl}}^2) = \hat{m}^2 \varphi^2/2,$$

$$f = \exp(2c \varphi), \quad \lambda_A = \log A, \quad \lambda_C = \log C_{\text{metric}}, \quad \theta = 2h_{\text{perp}} + h_{\text{parallel}}.$$

All ρ and p below are divided by $\mu^2 M_{\text{Pl}}^2$:

$$\rho_{\text{EM}} = f(E_{\text{cal}}^2 + B_{\text{cal}}^2)/2,$$

$$\rho = v^2/2 + U + \rho_{\text{EM}},$$

$$p_{\text{perp}} = v^2/2 - U + \rho_{\text{EM}}, \quad p_{\text{parallel}} = v^2/2 - U - \rho_{\text{EM}}.$$

The complete first-order system is

$$\lambda_A' = h_{\text{perp}}, \quad \lambda_C' = h_{\text{parallel}}, \quad \varphi' = v,$$

$$v' = -\theta v - \hat{m}^2 \varphi - c f(B_{\text{cal}}^2 - E_{\text{cal}}^2),$$

$$E_{cal}' = -(2h_{perp} + 2cv)E_{cal}, \quad B_{cal}' = -2h_{perp} B_{cal},$$

$$h_{perp}' = (\lambda - p_{parallel} - 3h_{perp}^2)/2,$$

$$h_{parallel}' = \lambda - p_{perp} - h_{perp}' - h_{perp}^2 - h_{parallel}^2 - h_{perp} h_{parallel}.$$

The algebraic Hamiltonian constraint is

$$C_g = h_{perp}^2 + 2h_{perp} h_{parallel} - \rho - \lambda = 0.$$

For regular initial data with nonnegative ρ and λ , the initial isotropic choice $h_{perp} = h_{parallel} = \sqrt{(\rho + \lambda)/3}$ satisfies it. An initially isotropic state is not a claim that anisotropic magnetic/electric stress will keep it isotropic. Do not enforce the constraint by resetting $h_{parallel}$ during evolution: its free propagation is a test.

G3. Exact exchange and constraint propagation

For any diagonal homogeneous sector define

$$Q = \rho' + 2h_{perp}(\rho + p_{perp}) + h_{parallel}(\rho + p_{parallel}).$$

The equations give

$$Q_{EM} = (f'/2)(B_{cal}^2 - E_{cal}^2) = c \int v(B_{cal}^2 - E_{cal}^2),$$

$$Q_{scalar} = -Q_{EM}, \text{ hence } Q_{total} = 0.$$

The previously established geometric identity becomes

$$C_g' = -\theta C_g - Q_{total}.$$

It follows that constraint-satisfying initial data retain $C_g = 0$ in exact arithmetic. For arbitrary initial constraint residual, $C_g(\tau) = C_g(0) \exp[-\int \theta]$. This is an exact conservation/constraint statement for Model G, not a result about a substituted QED stress tensor.

G4. Independent formulation, analytic fixtures and falsifier

With $A(0) = 1$, the exact flux laws are

$$B_{cal} = B_{cal0} \exp(-2\lambda A),$$

$$E_{cal} = f_0 E_{cal0} \exp(-2\lambda A)/f.$$

An independent reduced ODE can eliminate both electromagnetic evolution equations using these laws; comparison with the full system detects wrong dilution or scalar-coupling factors. Use a distinct numerical method for that reference when practical.

Analytic fixtures:

- Vacuum de Sitter: $\varphi = v = E_{cal} = B_{cal} = 0$, positive λ , $h_{perp} = h_{parallel} = \sqrt{\lambda/3}$ constant, and both logarithmic scale factors linear in τ .
- Massless homogeneous scalar: $\hat{m} = \lambda = E_{cal} = B_{cal} = 0$, isotropic $h_0 = |v_0|/\sqrt{6} > 0$. With $d = 1 + 3h_0 \tau$, $h = h_0/d$, $v = v_0/d$, $\varphi = \varphi_0 + (v_0/(3h_0)) \log d$, and $\lambda A = \lambda C = \log(d)/3$.
- Constant coupling $c = 0$: both electromagnetic fluxes dilute as A^{-2} , and scalar/gauge direct exchange vanishes.

The wrong-model control deletes only the scalar force $-c \int v(B_{cal}^2 - E_{cal}^2)$ while retaining the scalar energy and the varying- f Maxwell equation. It has

$$Q_{wrong} = c \int v(B_{cal}^2 - E_{cal}^2).$$

Let $\text{volume} = \exp(2\lambda A + \lambda C)$ and integrate $I' = \text{volume} \cdot Q_{wrong}$ alongside the state. Then

volume $C_g + I = C_g(0)$

must hold even for that wrong model, while $C_g = 0$ must fail for a nondegenerate case. This separates physical inconsistency from a numerical integration defect. Cases with zero c , v or electromagnetic invariant at all times cannot discriminate the missing force and must not be the only controls.

The declared numerical domain should include finite arguments, positive finite f , nonnegative $m\hbar$ and λ for the selected expanding fixtures, and finite scale factors. Guard exponential overflow explicitly rather than returning nonfinite stresses. f approaching zero is an ill-conditioned gauge sector; the fact that an exponential is formally positive does not certify numerical or quantum weak coupling.

4. Why these two models cannot yet be merged into Einstein-QED

Model F has a fixed canonical finite mode list, fixed magnetic b and no covariant directional quantum stress. Model G has a self-consistent homogeneous metric and complete classical stress but no charged quantum matter. Their matching pattern is the common action-derived exchange between a gauge kinetic coefficient and a scalar. Their scalar normalizations, regulators and state contents are different. If the same canonical scalar were later used, the units would relate by $\phi = (e \mu \hbar / m) \varphi$, and $x = (e \mu \hbar / m^2) E \alpha \lambda$; these unit conversions alone do not identify the selected polynomial and exponential coupling functions or close the missing quantum sector.

The missing interface is still a common causal quantum action or equivalent locally covariant prescription producing all of J_q , ρ_q , $p_{q\perp}$, $p_{q\parallel}$ in a stated state and the applicable force Ward identity. A moving geometry changes frequencies, physical momenta, magnetic field and state preparation; differentiating fixed- b flat-space formulas after the fact does not derive those changes.

The reference literature on locally covariant Dirac fields must be read with its sectional restrictions. Its conserved-current result does not license replacing the restricted stress argument by an unrestricted strong-background stress closure. [Zahn, especially section 4](#).

5. Optional local QED bridge, deliberately not the strong-field replacement

A separate controlled low-energy benchmark can use the leading spinor Euler-Heisenberg local term

$$L(E, B) = (E^2 - B^2)/2 + \kappa [(E^2 - B^2)^2 + 7(E \cdot B)^2],$$

$$\kappa = 2 \alpha^2 / (45 m_e^4).$$

For a pure homogeneous magnetic field it gives

$$\rho = B^2/2 - \kappa B^4, \quad p_{\perp} = B^2/2 - 3\kappa B^4, \quad p_{\parallel} = -\rho,$$

with $B_{\dot{}} = -2H_{\perp} B$. Inserting this stress into the same Einstein reduction conserves the homogeneous stress exactly for that truncated local action. One should independently vary the action to check these coefficients before implementation.

Its QED interpretation requires $|eB|/m_e^2 \ll 1$, slow variation on the electron scale and curvature components small relative to m_e^2 . The weak-field polynomial is not authorized at $B \gg B_{\text{QED}}$, and taking E of order the critical electric field requires treatment of real-time pair creation beyond this local real polynomial. These limitations concern the truncation written here; the full constant-field one-loop effective action has a wider field-amplitude range. [Dunne, sections on the constant-field action and its expansions](#).

The optional local branch earns a QED label only in its declared overlap regime. Neither that benchmark nor the scalar addition fills the full in-in current/stress/noise and ultraviolet completion gaps.

Appendix B - advisor decisions and next experiments

9 September 2026. Root synthesis of the completed advisor contract, independent skeptical equations and renormalization review. The five delegated agents reached a usage limit before a final implementation review. Their completed notes are preserved; the root finished the implementation audit, validation, proof graph and delivery. This document does not impersonate an unfinished advisor review.

Decision

Introduce a canonical scalar with a specified gauge-kinetic function and potential. First verify its energy exchange in the finite quantum-mode model. Separately verify its complete classical stress in a homogeneous evolving metric. Keep the reference-subtraction lemma as a distinct matching candidate. Attempt a connection only after the shared quantum current, energy and both directional pressures have been derived from one causal construction.

The objective is a verifiable closure in a declared regime. A full ultraviolet completion of Einstein-QED, the disappearance of the physical Landau pole, a singularity-resolution theorem, and experimental discovery of a new scalar are not established by this task.

Advisor down-selection

Candidate	Required new input	Practical test	Decision
Arbitrary bounded extra constant	None beyond changing the old model	Cofinal-divergence lemma	Cannot restore the uniform gap under the theorem's assumptions
Regulator-dependent matching coefficient	Fixed reference condition and common action variation	Exact continuous-integral subtraction identity	Keep as an algebraic candidate; physical matching remains open
Prescribed time function $\chi(t)$	External support work	Two distinct signed work defects	Do not label it a closed energy sector
Finite gauge-scalar quantum-mode model	Scalar equation, initial data and energy	Independent spinors, work balance and missing-term control	Implemented and checked
Classical scalar-Maxwell-Einstein model	Metric initial constraint and complete classical stress	Flux reduction, two integrators, exact fixtures and Ward defect	Implemented and checked
Strong-field semiclassical merged system	Common quantum current/stress, state and counterterms	Independent force Ward identity and regulator studies	Open; no sufficient rule admitted
Weak-field local Euler-Heisenberg gravity	Valid overlap regime and action-derived NLED stress	Low-field expansion and constraint propagation	Specified as a later benchmark; not run as a strong-field substitute

Four technical work packages

1. **Classify every freedom.** N and K are regulator labels; nK is quadrature resolution; μ_{RG} is a subtraction scale; ϕ is a new degree of freedom. Give each its equation or matching prescription. Rename the interpolation index used in the renormalization notes if implemented: its ν is not the finite scalar mass ratio ν .
2. **Close energy before coupling geometry.** Derive both scalar force and Maxwell exchange from the same action. Preserve the fixed magnetic energy's scalar dependence even when the magnetic field is held constant. Test zero frequency and the invariant zero-scalar branch separately from nonzero exchange.
3. **Close the classical stress independently.** Evolve both spatial Einstein equations without resetting the Hamiltonian constraint. Compare conserved-flux recovery and a separately written skeptical formulation. Distinguish shared-RHS comparisons from independent equations.
4. **Regress the full target into missing calculations.** Ask what quantum functional, state, subtraction, matching and estimates would suffice. Use the finite reviewed graph to find which obligations are actually met. Keep proposed connecting edges out of executable proof rules until justified.

Prioritized next experiments

P1. Common causal quantum stress in prescribed axial geometry

Start with a prescribed smooth axial metric and one specified positive gauge-kinetic function. Derive the tetrad Dirac equation, evolving physical longitudinal momentum and magnetic/Landau basis. Specify an admissible initial two-point state. Compute the quantum current and both directional stresses from the same regulated mode representation and allowed local subtraction.

Forward input: the present exchange signs and independently tested geometric constraint identity. Backward requirement: a quantum force Ward residual calculated from independently evaluated current and pressures. The bridge is accepted only if this residual tends to zero with independently varied time, quadrature and subtraction errors. Do not define a pressure by solving the Ward identity being tested.

Reject when the state is unspecified, a fixed-b basis is reused in changing B without its extra terms, or the current and stress use incompatible finite counterterms. Deliver the operator equations and regulator contract before writing a coupled Einstein integrator.

P2. Reference subtraction at fixed physical normalization

Use one physical normalization condition at a fixed reference field, state and momentum prescription. Derive all corresponding action variations. Study the combined current $S-C x_{\text{prime}}+D x^2$, energy and pressures, not just the separated C coefficient. Vary N and K along multiple cofinal paths and separately converge nK.

Reject a convergence claim if changing N,K also retunes scalar initial data, changes the measured charge, or changes the physical source. An unbounded bare coefficient is not automatically forbidden, but the finite renormalized observables and Ward identities must be established.

P3. Weak-field covariant overlap benchmark

Use the local Euler-Heisenberg expansion only where field amplitudes, derivatives and curvature satisfy its low-energy truncation assumptions. Vary the action before inserting its anisotropic stress. Compare with the quantum calculation in that overlap. A failure tests the normalization or truncation; success does not validate critical electric fields or ultrastrong-field pair production.

P4. Self-consistent semiclassical evolution

Only after P1-P3, couple the common observables to geometry. Determine a controlled treatment of higher derivatives, constraints, renormalization freedom and state/noise effects. Monitor all independent conservation residuals, regulator errors and approximation estimates. A PINN may later approximate this specified system; it cannot supply missing constitutive equations.

Interface plan

Put a current-review summary at the top of research home, retaining the earlier review below. Give the finite scalar, gravity, cutoff and proof-search branches dedicated pages. Each has a plain explanation, complete equation contract, observed plots, falsifier and next obligation. Recorded parameter selectors only switch computed datasets; they must not imply a live cloud simulation. Raw CSV and the research archive are downloadable. Full notes open only when requested through the existing accessible document dialog. Unknown routes and missing plot data produce clear fallback messages.

Stop and acceptance criteria

Acceptance requires independent symbolic identities, nondegenerate wrong-model failures, analytic fixtures, real historical-baseline comparison, independent complex spinors, numerical refinement, semantic result inspection, source integrity and optimized-Python gates. These are scoped validation layers, not a count of independent votes about truth. After concrete defects are repaired and required checks pass, package the results and expose the remaining obligations rather than expanding an uncontrolled parameter scan.

Appendix C - primary references and reading scope

1. The renormalized locally covariant Dirac field

<https://arxiv.org/pdf/1210.4031>

Authors: Jochen Zahn

Version: v3, 18 October 2013

Inspection: Primary research; full-text targeted inspection

Inspected sections: Proposition 4.1; section 4.2

Relevance: Current conservation and its ambiguity; restricted nonperturbative stress argument.

Scope limit: Does not supply our strong-F current-and-directional-stress closure.

2. Conservation of the stress tensor in perturbative interacting quantum field theory in curved spacetimes

<https://arxiv.org/abs/gr-qc/0404074>

Authors: Stefan Hollands; Robert M. Wald

Version: 2004 preprint; 2005 publication

Inspection: Primary research; abstract/scope inspection

Inspected sections: Abstract and stated field class

Relevance: Perturbative scalar construction of compatible conservation conditions.

Scope limit: Not a charged Dirac Bianchi-I existence theorem.

3. QED in the Exact Renormalization Group

<https://arxiv.org/abs/2107.14012>

Authors: Yuji Igarashi; Katsumi Itoh

Version: 2021

Inspection: Primary research; abstract and equation-index search inspection

Inspected sections: Flow/master-equation scope; published equation 74 search result

Relevance: RG consistency and gauge constraints at a momentum cutoff; one-loop beta coefficient.

Scope limit: Not regulator independence of our finite-mode current/stress.

4. Electric charge renormalization to all orders

<https://arxiv.org/html/2101.05154v2>

Authors: Stefan Dittmaier

Version: 2021

Inspection: Primary research; introduction inspection

Inspected sections: Introduction: Thomson limit

Relevance: Physical normalization must be declared.

Scope limit: No new numerical charge measurement in this project.

5. Running couplings from adiabatic regularization

<https://arxiv.org/html/1812.05564v2>

Authors: Antonio Ferreira; Jose Navarro-Salas

Version: v2

Inspection: Primary research; sections I-V targeted inspection

Inspected sections: Arbitrary subtraction mass; current/stress scale changes; Dirac discussion

Relevance: Scale-invariant semiclassical equations require coordinated coupling and subtraction changes.

Scope limit: Their signature, absorbed charge and FLRW setup need translation; no automatic magnetized Bianchi-I implementation.

6. Fine-structure constant: Is it really a constant?

<https://link.aps.org/doi/10.1103/PhysRevD.25.1527>

Authors: Jacob D. Bekenstein

Version: 15 March 1982

Inspection: Primary research; publication record/abstract inspection

Inspected sections: Original variable-coupling framework

Relevance: Historical precedent for a physical scalar electromagnetic coupling.

Scope limit: No claim that its particular model is experimentally established.

7. Black holes and membranes in higher-dimensional theories with dilaton fields

<https://www.sciencedirect.com/science/article/pii/0550321388900065>

Authors: G. W. Gibbons; K. I. Maeda

Version: 1988

Inspection: Primary research; publication record/abstract inspection

Inspected sections: Dilaton/gravity/gauge field coupling

Relevance: Historical precedent for scalar-gauge-gravity models.

Scope limit: Does not prove our proposed scalar extension is QED.

8. Inflationary Universe with Anisotropic Hair

<https://arxiv.org/html/0902.2833v2>

Authors: Masa-aki Watanabe; Sugumi Kanno; Jiro Soda

Version: 30 April 2009

Inspection: Primary research; action and basic equations inspected

Inspected sections: Equation 1 and section II

Relevance: Homogeneous anisotropic gauge-scalar benchmark; their f squared equals our f .

Scope limit: No quantum Dirac stress or joint-cutoff theorem.

9. Quantum Statistical Field Theory in Gravitation and Cosmology

<https://arxiv.org/abs/gr-qc/9403061>

Authors: B. L. Hu

Version: 1994

Inspection: Author research review; CTP passage inspected

Inspected sections: Causal backreaction and in-in effective action

Relevance: State-dependent causal backreaction requires the appropriate effective functional.

Scope limit: Does not calculate our state-dependent current or stress.

10. Covariant non-local action for massless QED and the curvature expansion

<https://arxiv.org/abs/1507.06321>

Authors: John F. Donoghue; Basem Kamal El-Menoufi

Version: 22 July 2015

Inspection: Primary research; abstract and full-text targeted inspection

Inspected sections: Nonlocal logarithm/curvature expansion scope

Relevance: A local number cannot replace a general nonlocal polarization structure.

Scope limit: Massless curvature expansion is not a massive ultrastrong-field solution.

11. QED vacuum polarization in a background gravitational field and its effect on the velocity of photons

<https://link.aps.org/doi/10.1103/PhysRevD.22.343>

Authors: I. T. Drummond; S. J. Hathrell

Version: 15 July 1980

Inspection: Primary research; publication record/abstract inspection

Inspected sections: One-loop curvature-dependent photon effective action

Relevance: Curvature operators have a regime and coefficient derivation.

Scope limit: Not an unrestricted nonperturbative Einstein-QED closure.

12. Faster than Light Photons in Gravitational Fields II - Dispersion and Vacuum Polarisation

<https://arxiv.org/abs/gr-qc/0203034>

Authors: G. M. Shore

Version: 2002

Inspection: Primary research; abstract/scope inspection

Inspected sections: Dispersion beyond lowest derivative treatment

Relevance: Derivative/frequency control matters.

Scope limit: No direct solver for the present joint strong-field geometry.

13. Vacuum birefringence and the Schwinger effect in (3+1) de Sitter

<https://arxiv.org/abs/1809.08977>

Authors: Mariona Banyeres; Guillem Domenech; Jaume Garriga

Version: v3, 9 June 2025

Inspection: Primary research; revised abstract/scope inspection

Inspected sections: Local polarization and negative-current interpretation

Relevance: Provides a comparison baseline for competing de Sitter current prescriptions.

Scope limit: Not treated as a settled resolution for a different background.

14. Classical constant electric fields and the Schwinger effect in de Sitter

<https://arxiv.org/html/2508.14973v2>

Authors: Mar Bastero-Gil; Paulo B. Ferraz; Antonio Torres Manso; Lorenzo Ubaldi; Roberto Vega-Morales

Version: v2, 17 April 2026

Inspection: Primary research; sections 1, 2, 2.1-2.3 inspected

Inspected sections: Dynamical supported constant E; matching; explicitly fixed metric and zero B

Relevance: Newer comparison prescription; source dynamics and matching alter the question.

Scope limit: Their geometry/stress are nondynamical, so this is not our Einstein-QED closure.

15. The Renormalization Group - Advanced Quantum Field Theory notes

<https://www.southampton.ac.uk/~doug/qft/aqft4.pdf>

Use: RG scale versus physical time; one-loop QED factor

Read scope: Sections 4.1 and 4.2, QED beta coefficient and scale dependence

Limits: Perturbative scale framework; not a continuum construction or a time-dependent coupling action

16. Einstein-Maxwell-scalar black holes with massive and self-interacting scalar hair

<https://arxiv.org/html/2003.01045v3>

Use: Established scalar-dependent gauge action model class; independent action normalization used here

Read scope: Sections 1-2 action, equations, stress, tests and model classes

Limits: Spherical charged-black-hole results not imported into Bianchi quantum model

17. Well-posed formulation of Einstein-Maxwell effective field theory

<https://arxiv.org/html/2112.05603v1>

Use: Well-posedness and EFT domain must be specified

Read scope: Abstract, section 3.6 weak coupling, final strong-hyperbolicity statement

Limits: Does not establish arbitrary strong-coupling or semiclassical quantum closure

18. Purely Electric, Magnetic, and Dyonic Black Holes in Einstein-Euler-Heisenberg Theory

<https://arxiv.org/html/2607.21938v1>

Use: Recent connected NLED branch and scope audit

Read scope: Abstract, introduction, selected magnetic/horizon discussion

Limits: No full numerical reproduction; polynomial strong-field applicability not established by citation

19. Heisenberg-Euler Effective Lagrangians: Basics and Extensions

<https://arxiv.org/abs/hep-th/0406216>

Use: Relevant effective-action reference with reading status exposed

Read scope: Abstract and bibliographic record only this round

Limits: Not claimed fully read this round; no imported coefficient theorem